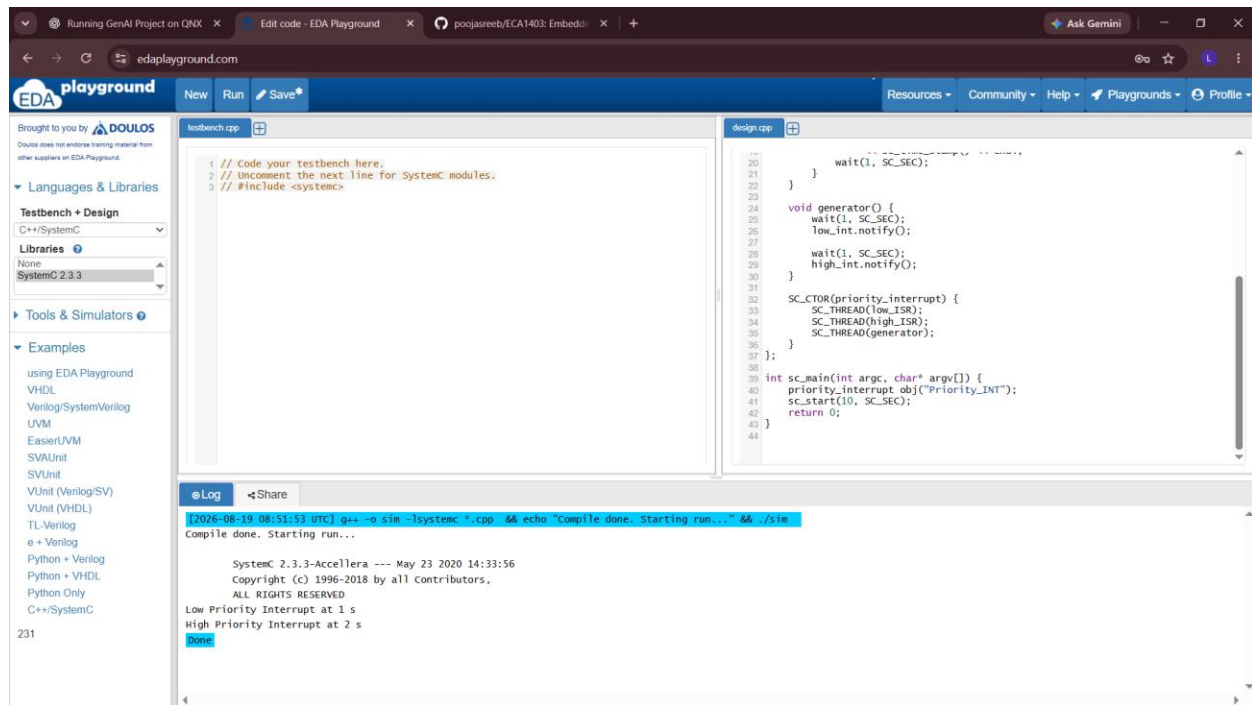




SOURCE CODE

```
#include <systemc.h>
```

```
SC_MODULE(priority_interrupt) {
    sc_event low_int, high_int;
```

```
    void low_ISR() {
        while (true) {
            wait(low_int);
            cout << "Low Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }
}
```

```
    void high_ISR() {
        while (true) {
            wait(high_int);
            cout << "High Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }
}
```

```
void generator() {
    wait(1, SC_SEC);
```

```
        low_int.notify();

        wait(1, SC_SEC);
        high_int.notify();
    }

    SC_CTOR(priority_interrupt) {
        SC_THREAD(low_ISR);
        SC_THREAD(high_ISR);
        SC_THREAD(generator);
    }
};

int sc_main(int argc, char* argv[]) {
    priority_interrupt obj("Priority_INT");
    sc_start(10, SC_SEC);
    return 0;
}
```